First-Divergence Consequence Analysis: Separating Quantization Onset and Output Discrepancy in Masked Visual Generators

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Abstract

A small local change in an iterative generator can cause either no sampled-token change or a mismatch that propagates to the output. First-Divergence Consequence Analysis (FDCA) separates these events through an explicit measurement contract: a common anchor state, a natural quantization seed drawn under a declared coordinatewise coupling, a common propagation kernel, audited random events, and inference over class–generation-seed blocks. The mathematical contribution consists of a stable-set additive-cost bound, an expectation identity for first-divergence paths, and a finite-horizon influence envelope under global hybrid-state assumptions. These results do not prescribe opposite trends in incidence and consequence and do not automatically bound LPIPS.

Under a fixed 32-step Halton schedule we compare one FP32-versus-W8 transition and propagate both branches with a common W8 suffix. In separate H-MaskGIT-T ($n = 48$) and H-MaskGIT-S ($n = 40$) CFG3 panels, moving from target unresolved fraction 0.3 to 0.5 lowers batch split probability while increasing both the equal-block split-conditional LPIPS and the mean of blockwise incidence-weighted LPIPS. Original paired-block intervals support conditional-LPIPS increases of 0.02000 and 0.02038, and incidence-weighted increases of 0.001714 and 0.001966. The latter is a mean of products, not a product of panel means. A labeled post-hoc CPU reanalysis checks aggregation, bootstrap robustness and schedule dependence. The commit batch shrinks from 14 to 8 positions; mean coordinate TV does not show the same decrease as batch split probability.

The contribution is an auditable separation of onset and conditional output discrepancy in this intervention. The observations establish neither a universal reverse-time law, persistent-quantization quality degradation, human preference, nor a validated precision-selection policy. Frozen source resolves per-output-channel quantization, including a documented deviation from the per-tensor plan; public compact evidence retains further reproducibility limits.

Revision and evidence boundary. This candidate corrects the interpretation of existing measurements and adds CPU reanalysis of public compact tables. The v1.0.0 PDF, source, preregistrations and frozen results remain historical records. No GPU inference, model training, new images or new perceptual measurements were performed. The DOI 10.5281/zenodo.21965747 identifies archived v1.0.0, not this unreleased revision.

1 Introduction

Local approximation diagnostics and endpoint metrics answer different questions. Logit total variation (TV) measures a change in a categorical law. A paired image distance measures discrepancy between two specified outputs. A coupling connects them by declaring a joint sampling experiment, including when the outputs remain equal and what happens after the first sampled-token mismatch.

FDCA makes that joint experiment explicit for fixed monotone masked decoding. At a common anchor, the reference and approximate categorical laws produce a natural seed under coordinatewise

maximal coupling. Both resulting states then use the same approximate kernel. This separates the probability of a batch split from the output consequence conditional on that split. Event-addressed random numbers, zero controls and block-level aggregation make the contract inspectable. The construction addresses a measurement question; it does not establish the optimal coupling for an image metric or the best deployment policy.

First disagreement is a classical stopping-time construction [Lindvall, 2002, Völlering, 2016, Ernst et al., 2019].

Single-point sensitivity interventions are also not unique to FDCA. In particular, Q-ARVD [Tang et al., 2026] studies frame-specific quantization sensitivity followed by a full-precision suffix; FDCA's natural-seed panels instead use a common W8 suffix. The distinction lies in the declared discrete onset law, conditional consequence, natural seed, coupling and RNG audits, and paired block inference together, rather than in the general observation that an early error can propagate.

The strongest empirical result is bounded but reproducible. Between target unresolved fractions 0.3 and 0.5, two CFG3 W8 checkpoint panels show $H \downarrow$, $M_{\text{equal}} \uparrow$, and $R \uparrow$, where H is mean batch split probability, M_{equal} is the equal-block conditional LPIPS, and R is the mean blockwise product. Batch size, state, coordinates, remaining horizon and seed composition all vary across anchors. The result therefore motivates measuring incidence and consequence separately; it does not identify a unique cause or imply the pattern holds at every anchor.

Contributions.

1. **A specified measurement framework.** A common anchor, one natural FP32-versus-W8 seed transition, common W8 propagation, coordinatewise coupling, semantic RNG and class–seed block inference define the measured discrepancy.
2. **Finite monotone-decoder theory.** We give a pathwise stable-set bound, a first-divergence expectation identity and a clipped DAG envelope. The envelope requires global state classes containing both branches and the hybrids used in the proof. Its finite inverse uses nilpotency, not contraction.
3. **Two checkpoint panels with a supported contrast.** In the stated CFG3 W8 intervention, lower batch split probability at target 0.5 coexists with higher equal-block conditional and incidence-weighted LPIPS than at 0.3. Matched timing, CFG1 and saturated W4 panels have distinct supporting roles.
4. **An accuracy and reproducibility correction.** We distinguish $\mathbb{E}_b[hm]$, $\mathbb{E}_b[h]\mathbb{E}_b[m]$, and the split-weighted population conditional mean; reproduce legacy numerical checks; and provide labeled post-hoc diagnostics, input hashes, generated tables and figures, and a buildable revised manuscript.

2 Related Work

2.1 Masked generation and schedule dependence

MaskGIT introduced iterative bidirectional visual-token generation [Chang et al., 2022]. Halton scheduling uses a spatially dispersed sequence [Halton, 1960, Besnier et al., 2025]. Work on factorized masked-diffusion approximations and inference schedules analyzes bias caused by simultaneous unmasking [Park et al., 2025, Lavenant and Zanella, 2025, Chen et al., 2026]. The MaskGIT sampler analysis of Hayakawa et al. [2026] studies its implicit temperature and order-selection mechanism. These results make the schedule part of the model of the experiment, not a neutral time label. FDCA fixes a branch-independent schedule and studies the declared paired implementation intervention; changing the schedule requires a new contract.

2.2 Coupling and comparison arguments

Maximal agreement and MEXIT ask how long stochastic processes can remain equal [Völlering, 2016, Ernst et al., 2019]. Coordinatewise maximal coupling minimizes one coordinate's mismatch probability.

It does not generally maximize whole-batch agreement or minimize a terminal LPIPS distance. Dobrushin comparison arguments bound discrepancy through local conditional perturbations [Rebeschini and van Handel, 2014]. Our finite, time-ordered influence matrix follows this lineage. The monotone commit order makes it nilpotent; measured neural local profiles remain different from the global suprema required for a certificate.

2.3 Visual quantization and error propagation

Q-ARVD [Tang et al., 2026] quantizes one autoregressive video frame while keeping the prefix and subsequent frames full precision in its sensitivity experiment (Section 3.1, Equation 3). This is a close precedent for localized shock/propagation analysis. FDCA instead defines a categorical natural-seed split and propagates both branches using a common W8 suffix. Neither a one-time perturbation nor the observation of error propagation is an exclusive novelty claim here.

For continuous diffusion, Liu et al. [2026a] analyze accumulated quantization error through a first-order propagation decomposition and develop TCEC compensation. That local linearization and continuous denoising setting differ from FDCA's categorical split law and its monotone discrete-state contract. PTQ4ARVG [Liu et al., 2026b] develops quantization/calibration methods for autoregressive visual generation, and Shift-and-Sum [Moon and Ham, 2026] targets attention-value reconstruction and calibration across scales. Caching and mixed quantization also seek computational savings [Yan et al., 2025, Wu et al., 2025].

FDCA's contribution is the combined onset/consequence measurement contract, not demonstrated superiority over these quantizers or sensitivity methods. The present data do not show that it predicts outcomes or selects precision better than logit TV, seed geometry, or simple early protection. Those comparisons require held-out policy tests (Section 9).

2.4 Output metrics

LPIPS [Zhang et al., 2018] and DINOv2 embedding distance [Oquab et al., 2024] are paired output discrepancies. A ResNet-50 classifier [He et al., 2016] supplies secondary diagnostics. A greater distance between the two specified outputs establishes neither a consistent quality loss from a reference distribution nor human detection or preference.

3 Setup, Estimands and Aggregation

3.1 Fixed monotone masked decoder

Let I be a finite set of N token positions, V a finite codebook, and $\mathtt{M} \notin V$. A branch-independent schedule is an ordered partition $C = (C_1, \dots, C_K)$ of I, fixed independently of either branch's realized tokens. Positions in C_{k+1} are committed at transition $k+1$. Committed token values are never revised. At a pre-transition state x_k, reference and approximate batch laws factorize as

$$P_k(\cdot \mid x_k) = \bigotimes_{u \in C_{k+1}} p_{k,u}(\cdot \mid x_k), \qquad Q_k(\cdot \mid x_k) = \bigotimes_{u \in C_{k+1}} q_{k,u}(\cdot \mid x_k).$$

This product assumption, with independent coordinate coupling events conditional on the state, is needed for the split-probability product formula. It is not needed for the stable-set argument. Any additional memory affecting transitions must be part of the state; token grids alone are insufficient for a model with branch-dependent hidden memory or caches.

Definition 3.1 (First divergence)**.** For two coupled processes starting equal, let $\tau = \inf\{k \geq 1 : X_k \neq Y_k\}$, with $\tau = \dagger$ if no split occurs. We use $G_k = \{\tau > k\}$ for survival without a prior split, treating $\tau = \dagger$ as surviving every finite stage.

The terminal Hamming cost is $d_\lambda(x, y) = \sum_u \lambda_u \mathbf{1}\{x_u \neq y_u\}$, where $\lambda_u \geq 0$ and $\sum_u \lambda_u = 1$.

Table 1: Frozen 32-step, 16×16 Halton schedule. Transition numbers are one-indexed. The target 0.1 anchor is the final flush of 41 remaining tokens.

Target ρ	Transition	Unresolved	Attained ρ	c
0.1	32	41	0.16015625	41
0.3	30	72	0.28125	14
0.5	24	126	0.4921875	8
0.7	16	177	0.69140625	6
0.9	6	231	0.90234375	5

3.2 Unresolved fraction and schedule

The label $\widetilde{\rho}$ is a *target unresolved-token fraction*, not a quantization strength or a generation-step number. Larger fractions occur earlier in decoding. The nearest audited pre-transition state has attained fraction ρ_{att}, which is generally different from the target.

3.3 Block estimands and population conditional means

A block b denotes a class–generation-seed pair. Within one panel and anchor r, let S denote a split of the single anchor transition and define

$$h_{b,r} = 1 - \prod_{u \in C(r)} [1 - \mathrm{TV}(p_u, q_u)], \tag{1}$$

$$m_{b,r} = \mathbb{E}_\Gamma[d \mid S, b, r], \qquad R_{b,r} = h_{b,r} m_{b,r}. \tag{2}$$

The value $h_{b,r}$ is the conditional batch split probability under the specified coordinate-independent coupling at the common state. It is computed from the categorical laws (up to floating-point evaluation), not from a count of sampled split events. Its across-block uncertainty is therefore not a binomial Wilson interval. The conditional mean $m_{b,r}$ is estimated from repeated conditional rollouts and retains Monte Carlo error. For LPIPS, d denotes the distance between the two decoded terminal grids.

For the population that first chooses a block with equal weight, define

$$H = \mathbb{E}_b h_{b,r}, \qquad M_{\mathrm{equal}} = \mathbb{E}_b m_{b,r}, \qquad R = \mathbb{E}_b[h_{b,r} m_{b,r}], \qquad M_{\mathrm{split}} = R/H \quad (H > 0). \tag{3}$$

Here M_{equal} is the *equal-block* conditional mean. In contrast, M_{split} is the conditional mean after drawing an equal-weight block and conditioning that population on a split; this reweights blocks in proportion to h. The historical tables called M_{equal} "conditional LPIPS." Throughout this revision that phrase is qualified as equal-block conditional LPIPS. In general,

$$R = H M_{\mathrm{equal}} + \mathrm{Cov}_b(h, m), \qquad \mathrm{Cov}_b(h, m) = n^{-1} \sum_b (h_b - H)(m_b - M_{\mathrm{equal}}). \tag{4}$$

Thus multiplying printed panel means is not the published incidence-weighted estimator. The covariance uses population normalization $1/n$.

If $h_{b,r} = 0$, its contribution to R is zero, but a conditional mean on that zero-probability event is unobserved, not an observed zero. An all-block M_{equal} then requires an explicit domain convention and cannot be manufactured by zero filling; report the positive-h domain instead. If $H = 0$, M_{split} is undefined. All reported primary data have positive h.

3.4 One shock and persistent approximation are different estimands

The natural-seed panels use a common FP32 prefix state, one FP32-versus-W8 transition, and a common W8 kernel on both subsequent branches. Equal branches remain equal under the shared self-coupling. Conditional on a seed split, the suffix measures state-mediated propagation without fresh same-state

FP32-versus-W8 innovation. Persistent approximation would instead compare an FP32 branch and a W8 branch throughout the suffix and include further innovation. The current $h_{b,r}$ values do not condition on survival along those persistent paths. Summing them or multiplying their complements across anchors does not estimate a persistent first-divergence CDF or its total expected discrepancy.

4 Theory and Its Scope

4.1 Stable-set additive-cost bound

Definition 4.1 (Future-stable set)**.** At a common state s before transition $k+1$, $F_k(s)$ contains coordinates equal in both branches that no allowed continuation of either branch revises.

Theorem 4.2 (Stable-set bound)**.** *Assume the finite common monotone schedule and equal initial states above. Let $d(x, y) = \sum_u \lambda_u c_u(x_u, y_u)$ with finite nonnegative λ_u, nonnegative coordinate costs, $c_u(a, a) = 0$, and bounded $\mathrm{diam}(c_u) = \sup_{a,b} c_u(a, b) < \infty$ over the terminal domain. Then on $\{\tau \neq \dagger\}$,*

$$d(X_K, Y_K) \leq \sum_{u \notin F_{\tau-1}(X_{\tau-1})} \lambda_u \mathrm{diam}(c_u),$$

and the terminal cost is zero on $\{\tau = \dagger\}$. Taking expectations gives the corresponding expected-cost bound.

Proof. Immediately before the first split both states agree. Every coordinate in the future-stable set remains equal through all allowed continuations, so its terminal cost is zero. Bound each remaining nonnegative coordinate cost by its finite diameter, multiply by its nonnegative weight, and sum. On no split, all terminal coordinates agree and every diagonal cost is zero. □

For uniform Hamming cost this is the fraction of positions still capable of changing. It is often loose early in decoding. Signed costs and negative weights are outside the stated theorem. LPIPS is not a coordinate-additive token cost; a decoder-to-metric bound with valid constants would be needed to transfer this bound to LPIPS, and none is established here.

4.2 Full-path expectation identity

For a specified coupled *full-path* law, define

$$h_k(s) = \mathbb{P}(\tau = k + 1 \mid G_k, X_k = Y_k = s), \quad m_k(s) = \mathbb{E}[d(X_K, Y_K) \mid G_k, X_k = Y_k = s, \tau = k + 1].$$

Proposition 4.3 (Onset–consequence identity)**.** *For an integrable terminal cost with zero cost on $\{\tau = \dagger\}$,*

$$\mathbb{E}[d(X_K, Y_K)] = \sum_{k=0}^{K-1} \mathbb{E}[\mathbf{1}\{G_k\} h_k(X_k) m_k(X_k)].$$

Zero-hazard terms contribute zero without assigning an observed conditional mean.

Proof. The disjoint events $\{\tau = k + 1\}$ partition the event of a split. Expand the expected cost over them and apply conditional expectation first on G_k and the common state, then on the split event. The no-split contribution is zero by assumption. □

This is the law of total expectation. It implies no sign relationship between h and m. The one-anchor single-shock identity $R_{b,r} = h_{b,r} m_{b,r}$ is a different specialization: it has one allowed innovation transition, not a sequence of persistent hazards. Mixing the anchor interventions does not recover the survival-conditioned quantities of the proposition.

4.3 Clipped innovation–influence DAG envelope

At stage k, let $\mathcal{S}_k$ contain every possible state of both branches and every intermediate state obtained by changing previously committed coordinates one at a time between them. Unresolved coordinates and the stage are fixed consistently in this class. For $u \in C_{k+1}$, define global suprema

$$s_u = \sup_{x \in \mathcal{S}_k} \mathrm{TV}(p_{k,u}(\cdot \mid x), q_{k,u}(\cdot \mid x)), \qquad M_{uv} = \sup_{\substack{x,y \in \mathcal{S}_k \\ x \sim_v y}} \mathrm{TV}(q_{k,u}(\cdot \mid x), q_{k,u}(\cdot \mid y)),$$

where v is in a strictly earlier commit block and $x \sim_v y$ means equality except at v. Set $M_{uv} = 0$ for other pairs.

Theorem 4.4 (Clipped DAG envelope)**.** *Assume the fixed branch-independent monotone schedule, batch laws determined by the token grid and stage, compatible product marginals, global hybrid-state coverage above, and conditional coordinatewise maximal coupling of $p_{k,u}(\cdot \mid X_k)$ with $q_{k,u}(\cdot \mid Y_k)$ at every commit. In commit order set*

$$\bar{r}_u = \min\{1, s_u + \sum_{v<u} M_{uv} \bar{r}_v\}.$$

Then $\mathbb{P}(X_K[u] \neq Y_K[u]) \leq \bar{r}_u$ and $\mathbb{E}[d_\lambda(X_K, Y_K)] \leq \lambda^\top \bar{r}$.

Proof. Write π_u for terminal mismatch probability at u. Monotonicity makes it equal to mismatch probability at that coordinate's commit. Maximal coupling and the TV triangle inequality give a same-state term at most s_u and a state-change term. Interpolate from X_k to Y_k through the assumed hybrids. Changing an earlier mismatched coordinate v contributes at most M_{uv}; a coordinate that agrees contributes zero. Taking expectations bounds the state term by $\sum_{v<u} M_{uv}\pi_v$. Since probabilities are at most one, $\pi_u \leq \min\{1, s_u + \sum_{v<u} M_{uv}\pi_v\}$. Induction in commit order gives the asserted coordinate bounds. Sum with nonnegative Hamming weights to obtain the terminal bound. □

The matrix is strictly block-triangular with K commit blocks, so $M^K = 0$ and $(I - M)^{-1} = \sum_{j=0}^{K-1} M^j$. No contraction or spectral-radius assumption is required. This is a finite-horizon statement, not infinite-time stability.

Remark 4.5 (Measured profiles and metric transfer). Finite local measurements, q95 profiles and calibration maxima are not global supremum certificates. T0 enumerates the finite synthetic state classes; it does not validate these assumptions for arbitrary neural states. P05R's neural envelope was empirically vacuous and P1M found limited incremental within-anchor prediction. Neither these profiles nor observed LPIPS trends prove a neural LPIPS bound. Confidence samplers that rewrite token values, continuous diffusion, and new architectures require separate assumptions and definitions.

5 Implementation and Measurement Contract

5.1 Generator and state

The official Halton-MaskGIT source is pinned to `f61b0a1314717004dc7487531fd16a8bb71e1888` [Besnier et al., 2025]. H-MaskGIT-T uses hidden width 384, depth 6 and six attention heads; H-MaskGIT-S uses width 512, depth 12 and six heads. Both have a 16×16 grid and use 32 Halton steps with `randomize=False`. The Small asset ledger pins `ffc7fd4c5fc7b16010acc4aa342310484d7de62a`, 832,274,338 bytes and SHA-256 `760b9937d16157e7e518fbf405efe281552933242b552d5e8262cb10507fb0e2`. These are recorded provenance facts; the compact CPU analysis does not load or independently rehash that external checkpoint. The confidence sampler was inspected and excluded from the theorem because committed values can be rewritten even when mask occupancy is monotone.

5.2 Approximation provenance

Frozen runner snapshots identify deterministic symmetric *per-output-channel* round-to-nearest (RTN) quantization of eligible transformer linear weights. The imported `LinearWeightHomotopy` class constructs W4/W8 weights with `rtn_per_output_channel`; the function reduces the weight maximum over input dimension 1 while retaining that dimension, so its scale tensor has shape [out, 1]:

$$Q_b(W_{o,:}) = \mathrm{clip}\big(\mathrm{round}(W_{o,:}/\Delta_o), -q_{\max}, q_{\max}\big)\,\Delta_o, \qquad \Delta_o = \max_j |W_{o,j}|/q_{\max}.$$

All-zero rows return zero directly. PyTorch deterministic `round` uses ties to even and the zero point is zero. W8 has $q_{\max} = 127$. These are weight-only fake-quantized weights used in FP32 computation; activations, norms, embeddings, biases and the VQ decoder remain FP32.

The historical P2P preregistration instead names `per_tensor_symmetric_RTN`. This is a preserved plan-to-source deviation; its YAML is not silently rewritten. Matching frozen P2P/P3S runner imports and code-manifest hashes to the public quantizer source resolves the source-level granularity. This revision does not load models or independently regenerate historical runtime scale tensors or quantized-weight hashes. The sanitized receipt is `docs/evidence/QUANTIZER_SOURCE_RECEIPT.json`. No packed-kernel runtime, speedup or VRAM saving is measured by this output-fidelity experiment. W4 is retained only as saturated stress evidence.

5.3 Coordinate coupling and split law

For categorical laws p, q, draw $x \sim p$, accept $y = x$ with probability $\min\{1, q(x)/p(x)\}$, and otherwise draw from $(q-p)_+/\mathrm{TV}(p,q)$. If $\mathrm{TV}(p,q) = 0$, return (x,x) without normalizing a zero residual. Then $X \sim p$, $Y \sim q$, and $\mathbb{P}(X \neq Y) = \mathrm{TV}(p,q)$. Independently addressed coordinate events yield Equation (1) for a factorized commit batch conditional on its common state. Coordinate maximality is not batch, path or LPIPS optimality. The coupling choice changes the paired estimand, even when both marginal laws are unchanged.

5.4 RNG and logging

Semantic keys identify experiment, cell, block, anchor, replicate, attempt, step, position, event kind and draw. Separate event kinds address reference proposals, accept/reject events, residual draws and suffix events. Registries classify intentional shared reference streams and reject unclassified duplicate or missing keys. P2B retains equal as well as mismatched selected-token pairs; the historical reports find no selected MASK or clamp-induced mismatch. Zero-seed controls check equal terminal grids, images and metrics. These are historical execution/audit findings, not newly rerun generation in this revision.

LPIPS uses the AlexNet backbone, DINOv2 uses ViT-S/14 cosine distance, and secondary P2P diagnostics include classifier JSD, SSIM and pixel L_1. A smoke numerical audit selected scalar metric inference after a scalar-versus- batch tolerance failure. The compact data retain scores; they do not independently recreate checkpoint loading, image decoding, feature extraction or human ratings.

6 Experimental Design and Inference

6.1 Panels and statistical unit

Within each panel, target 0.3 and 0.5 use the same block identifiers. CFG1 uses a 16-block subset of the Small primary panel; W4 uses a 24-block subset of the Tiny W8 panel. They are sensitivity/stress comparisons, not independent extra replications. The compact files permit checking panel IDs, counts and matching. They omit a complete raw class-ID/base-seed manifest, so this reanalysis does not independently certify class–seed uniqueness or predecessor disjointness. P2P compact CSVs omit CFG metadata; CFG3 is sourced from the preserved protocol rather than independently filtered from those rows. Repeated seeds within a class are not automatically pseudoreplication; the block bootstrap assumes

Table 2: Panel contracts. A block is one class–generation-seed pair. Conditional replicates, anchors, coordinates and output images are not additional independent samples. Overlap is assessed within a checkpoint, never by matching Tiny and Small numeric block identifiers.

Panel	CFG	Blocks	Anchors	Replicates
P2P Tiny W8 primary	3	48	5	8
P3S Small W8 primary	3	40	2	8
P3S Small W8 sensitivity	1	16	2	4
P2P Tiny W4 stress	3	24	2	4
P2P matched timing	3	24	2	2
P3S matched timing	3	40	2	2

blocks are independent for the chosen class–seed population, with no separately assessed class-cluster effect. A class-cluster reanalysis is unavailable without the true class manifest.

6.2 Selection history and matched timing

P2P follows the P2N observations, after which the 0.5 versus 0.3 contrast was specified and the execution records describe new blocks. P3S records 40 new blocks and a conservative power plan using half the P2P paired effect, candidates 32/40/48, and a 90% target; the archived audit reports 40 as the first adequate candidate. These are preserved historical planning records. Public YAML and embedded timestamps alone do not provide independent external proof that the analysis was registered before outcome access.

Matched timing retains the same seed coordinate and token substitution across two anchor contexts according to the original execution logs. Its purpose is control of seed identity. State, remaining horizon and coordinate context still vary, so it is not a pure causal time intervention. Public compact matched files lack the full token/coordinate identity needed to revalidate that construction independently. Its score contrast is reproducible, with $n = 24$ for Tiny and $n = 40$ for Small, and is kept separate from natural-shock results.

6.3 Legacy inference and post-hoc supplement

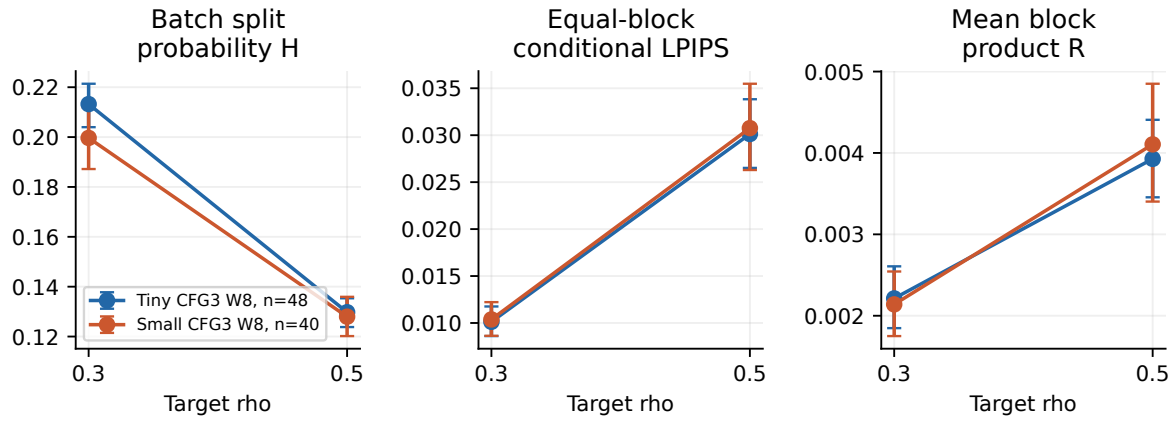
The headline conditional-LPIPS and incidence-weighted-LPIPS intervals retain the original 20,000-resample paired block bootstrap, semantic SHA-256 label seeds, and NumPy `higher` quantiles. Existing gate intervals for DINOv2 and matched LPIPS use that same original rule. Each original contrast resamples paired block differences, preserving the two anchors. Its semantic seed is contrast-specific; the new supplement instead shares block indices across metrics within each panel. Legacy reaggregation checks the eight stored contrasts and two stored OOF MAE ratios; checking both CI endpoints is stronger than the previous public script's lower-only checks.

A separately labeled *post-hoc robustness supplement* uses 100,000 paired block resamples and `linear` quantiles. Stable SHA-256 label-to-seed mapping, RNG, resample count, input hashes, Python/NumPy/SciPy versions and exact outputs are stored in the generated audit. Seeds are not based on process-dependent Python `hash()`. Different seeds can shift Monte Carlo endpoints; no endpoint is forced to match a comparison report. Ratios and M_{split} are recalculated inside each bootstrap sample. Paired t intervals, leave-one-block-out influence, trimmed means and blockwise signs are diagnostics, not randomized-treatment causal tests.

The supplement also reports 99.1667% marginal intervals for the six H, M_{equal}, R contrasts in the *two primary CFG3 W8 panels only*. Bonferroni accounting for this six-endpoint family does not correct all post-hoc hypothesis selection in the research program. CFG1, W4, matched timing and interactions are outside that family. Conditional rollout counts are only eight (primary), four (sensitivity/stress) or two (matched) per block/anchor; block intervals do not separately identify all within-block Monte Carlo error.

Table 3: Primary CFG3 W8 block means and paired contrasts, $\Delta = .5 - .3$. M_{equal} is equal-block conditional LPIPS and $R = \text{mean}_b(hm)$. Conditional-LPIPS and R headline intervals use the original 20k/higher rule; H inference is supplied only in the post-hoc supplement.

Panel	Metric	n	$\rho = .3$	$\rho = .5$	Δ	Legacy 95% CI
Tiny CFG3 W8	H	48	0.213227	0.129756	-0.083471	–[†]
Tiny CFG3 W8	M_{equal}	48	0.010126	0.030126	0.020000	[0.017189, 0.022874]
Tiny CFG3 W8	R	48	0.002213	0.003927	0.001714	[0.001368, 0.002066]
Small CFG3 W8	H	40	0.199666	0.127957	-0.071710	–[†]
Small CFG3 W8	M_{equal}	40	0.010380	0.030756	0.020376	[0.017142, 0.023806]
Small CFG3 W8	R	40	0.002140	0.004106	0.001966	[0.001499, 0.002477]



Post-hoc 100,000 paired-block bootstrap, linear 95% intervals. Blocks are class-seed units; CFG3 W8 only.

Figure 1: Separate axes for H, M_{equal} and R in the primary CFG3 W8 panels; block counts are panel-specific. Figure intervals use the labeled post-hoc 100k/linear paired-block rule. Directions concern this one-shock intervention, not persistent approximation or image quality.

7 Results

7.1 Primary contrast: lower batch incidence and higher discrepancy

Table 3 gives the two primary panels. Both show $H \downarrow$, $M_{\text{equal}} \uparrow$, $R \uparrow$ from target 0.3 to 0.5. Legacy equal-block conditional LPIPS intervals are approximately $[0.01719, 0.02287]$ for Tiny and $[0.01714, 0.02381]$ for Small; incidence-weighted intervals are approximately $[0.001368, 0.002066]$ and $[0.001499, 0.002477]$. These are within-checkpoint paired contrasts. The exact source-linked values and sample counts are generated below. The three directions hold simultaneously in 43/48 Tiny and 39/40 Small blocks. This is a descriptive blockwise pattern, not a theorem. DINOv2 supports a positive equal-block conditional contrast in the original hypotheses, with absolute effects differing by checkpoint. Tiny's full five-anchor R profile is not strictly monotone: the $0.1 \to 0.3$ contrast is about -0.000054 with a post-hoc interval spanning zero. The final-flush anchor must therefore remain visible in the full-anchor supplement.

7.2 Aggregation and attribution

The equal-block and split-population conditional means differ because h and m covary. Tiny M_{split} is about $0.010378 \to 0.030267$; Small M_{split} is about $0.010716 \to 0.032086$. In Small, HM_{equal} is about 3.13% below R at target 0.3 and 4.14% below it at 0.5. These are covariance effects, not a rounding discrepancy. For the $0.5 - 0.3$ contrast the exact symmetric arithmetic decomposition is

Table 4: Aggregation check from block data. Population covariance uses $1/n$. $M_{\text{split}} = R/H$ is not the historical equal-block conditional column.

Panel	ρ	M_{equal}	M_{split}	R	HM_{equal}	Covariance
Tiny CFG3 W8	0.3	0.010126	0.010378	0.002213	0.002159	0.000054
Tiny CFG3 W8	0.5	0.030126	0.030267	0.003927	0.003909	0.000018
Small CFG3 W8	0.3	0.010380	0.010716	0.002140	0.002073	0.000067
Small CFG3 W8	0.5	0.030756	0.032086	0.004106	0.003935	0.000170

Table 5: Blockwise symmetric attribution and simultaneous blockwise directions. The identity is checked in the generated audit; the components are descriptive, not independent causal effects.

Panel	A_h	A_m	ΔR	Joint direction
Tiny CFG3 W8	-0.001743	0.003457	0.001714	43/48
Tiny W4 stress	-0.001887	0.023911	0.022024	21/24
Small CFG3 W8	-0.001511	0.003477	0.001966	39/40
Small CFG1 W8	-0.001742	0.003428	0.001686	15/16

$$A_h = \text{mean}_b \left[(h_{b,.5} - h_{b,.3}) \frac{m_{b,.5} + m_{b,.3}}{2} \right], \tag{5}$$

$$A_m = \text{mean}_b \left[(m_{b,.5} - m_{b,.3}) \frac{h_{b,.5} + h_{b,.3}}{2} \right], \qquad \Delta R = A_h + A_m. \tag{6}$$

Apply the identity per block and then average. Replacing the block expressions by products of panel averages omits covariance terms. Tiny contributions are approximately −0.001743 and +0.003457; Small contributions are approximately −0.001511 and +0.003477. This accounting is not a causal mediation analysis.

7.3 Schedule and local-TV diagnostics

The public expected seed count is $\sum_{u \in C} \text{TV}(p_u, q_u)$, so mean coordinate TV equals expected seed count divided by commit batch size c. This average does not reconstruct the full coordinate TV distribution. For $0.3 \to 0.5$, c decreases from 14 to 8. Tiny's mean coordinate TV is about $0.017015 \to 0.017232$ and its paired difference interval includes zero; this is not an equivalence result. Small's mean coordinate TV increases, approximately $0.015846 \to 0.016988$, with a positive paired difference interval. Batch h nevertheless falls in both panels. The product split law depends on how many coordinates can split as well as their TVs. Schedule/branching opportunity is therefore an explanation candidate, not an identified sole cause. The current observations do not isolate pure batch-size or time effects.

7.4 Matched timing, sensitivity and saturated stress

Matched timing gives a separate equal-block LPIPS contrast of about 0.025088 for Tiny ($n = 24$) and 0.022235 for Small ($n = 40$). The official CFG1 Small sensitivity panel has 16 overlapping blocks. Its M_{equal} and R contrasts have the same signs as CFG3; the paired CFG1-minus-CFG3 interaction is computed only on their common subset. Directional consistency does not establish equal effect sizes. The Tiny W4 stress subset uses 24 blocks overlapping W8. Its R is about $0.029407 \to 0.051431$, while

Table 6: Commit opportunities and mean coordinate TV, calculated from the public expected-seed-count field. Intervals are post-hoc 100k/linear paired-block intervals for $\Delta = .5 - .3$.

Panel	$c_{.3} \to c_{.5}$	TV at .3	TV at .5	Δ	Post-hoc 95% CI
Tiny CFG3 W8	$14 \to 8$	0.017015	0.017232	0.000217	[-0.000374, 0.000833]
Small CFG3 W8	$14 \to 8$	0.015846	0.016988	0.001143	[0.000417, 0.001889]

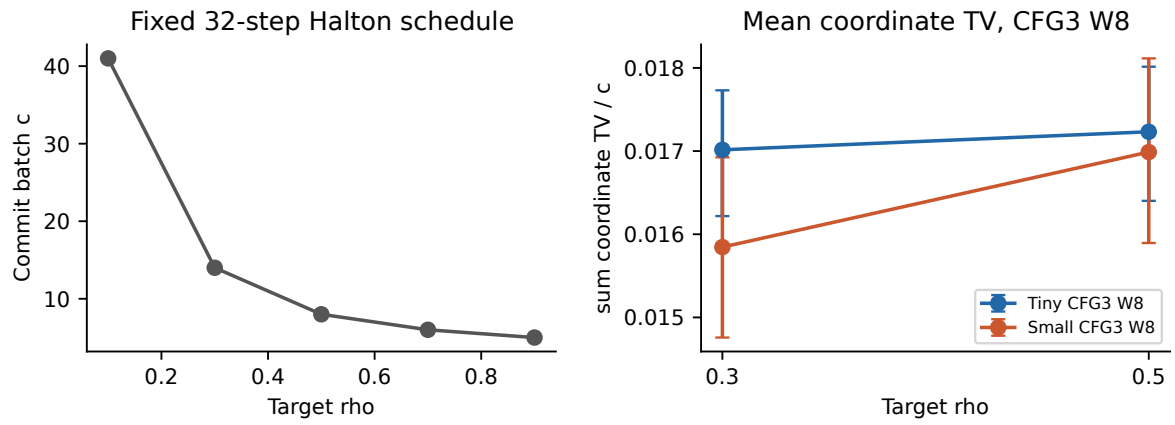


Figure 2: Batch schedule and mean-coordinate TV displayed separately. Falling batch split probability does not establish falling local quantization error. Remaining horizon, state, positions and seed composition also change.

Table 7: Matched-timing score contrasts are separate from natural-shock R. Tiny has 24 paired blocks, Small 40; each has two conditional replicates.

Panel	n	.3 LPIPS	.5 LPIPS	Δ	Legacy 95% CI
P2P matched	24	0.014954	0.040042	0.025088	[0.015525, 0.035879]
P3S matched	40	0.010036	0.032271	0.022235	[0.016028, 0.028862]

H is near saturation (0.9878 $\rightarrow$ 0.9367). It is an exploratory stress comparison; paired W4-versus-W8 comparisons use only their common subset. Neither subset is an independent non-saturated W8 replication.

7.5 Prediction evidence and its reproducibility limit

The archived analyses fit leave-one-block-out ridge models. A seed-only model uses anchor category, count, severity, surprisal and seed topology; adding post-suffix descendant features yields stored OOF MAE ratios of approximately 0.5166 (Tiny) and 0.5440 (Small). The CPU reproduction can reaggregate stored OOF errors. That is not an independent refit of the feature extraction and LOBO training pipeline. Descendant information is measured after propagation; it cannot be used as an online pre-terminal precision-selection feature. No current result establishes improvement over a strong prospective policy baseline at matched parameter bytes or measured cost.

Table 8: Exploratory paired differences in $\Delta = .5 - .3$, restricted to common within-checkpoint blocks. Same direction does not imply equal effect size.

Comparison	Metric	n	Difference in Δ	Post-hoc 95% CI
CFG1–CFG3	H	16	0.003193	[-0.011539, 0.017887]
CFG1–CFG3	M_equal	16	0.009261	[0.002095, 0.016495]
CFG1–CFG3	R	16	-0.000194	[-0.000860, 0.000471]
W4–W8 stress	H	24	0.027189	[0.007925, 0.045462]
W4–W8 stress	M_equal	24	0.004657	[-0.004832, 0.014864]
W4–W8 stress	R	24	0.020159	[0.011358, 0.030799]

8 Negative Findings and Limitations

P05R's finite-profile influence envelope was conservative but empirically vacuous in earlier regimes; P1M found little within-anchor incremental value. P2A remained incomplete at its historical stopping point; P2B subsequently closed the particular RNG/MASK/clamp audit questions in its own recorded stage. The revision preserves these outcomes and does not retroactively relabel P2A complete. Associations involving source age were unstable under adjustment, and no source-age causal claim is made. No discontinuous phase transition was fitted or tested.

The two checkpoints share an architecture family, VQ decoder and scheduler. The public data do not establish architecture, scheduler or coupling universality. A split hazard for persistent paths is different from these anchor-conditioned batch probabilities. LPIPS and DINOv2 measure specified output discrepancies; no human similarity, preference or quality study is included. Weight-only fake quantization supplies no packed-kernel speed or memory measurements.

The block bootstrap assumes independent class–seed blocks and addresses the empirical block population. Compact IDs do not reveal all class/seed provenance, predecessor overlap or potential class-level dependence. Repeating multiple seeds of a class is not by itself proof of pseudoreplication, and the original primary analysis already used block resampling. A class-cluster check needs the true manifest. Limited conditional rollout counts create additional Monte Carlo uncertainty. Historical audits are evidence about the runs they inspected; this CPU revision does not independently rerun their full model/metric pipeline, rehash excluded weights or validate every original token identity. Stored h and LPIPS/DINO scores are reaggregated, not independently reconstructed from per-coordinate categorical marginals or terminal images absent from these tables.

The historical per-tensor plan differs from frozen per-channel implementation source (Section 5.2); that deviation is now explicit. Exact historical runtime scale/weight tensor replay has not been performed. The theoretical influence envelope requires global coverage unavailable from finite neural local profiles, and no valid token-bound-to-LPIPS transfer constant is supplied.

9 Unexecuted Research Plan and Falsification Criteria

Every experiment in this section is a proposal. No new result or compute budget is implied by its inclusion.

P1: schedule and split opportunities. First test the same model in a 2×2 design: the existing batch sequence versus $c = 8$ for each of 32 steps, and Halton order versus a fixed permutation. Changing the schedule also changes prefix states, so it is not a pure causal intervention on c. A complementary diagnostic uses one identical anchor state and nested sets of $c = 1, 4, 8$ coordinates: compare p with q on those positions, use common p on all others, and then common q in the suffix. This is a diagnostic intervention distinct from the full natural quantization transition. Freeze comparisons before evaluation. If equal-c removes the incidence reversal, narrow the interpretation to schedule-dependent opportunities; if conditional/weighted differences vanish after seed count and severity are controlled, do not claim residual timing sensitivity.

P2: a different discrete generation structure. A first candidate is single-token AR generation such as LlamaGen, subject to separate verification of official checkpoint, tokenizer and runtime feasibility; no GPU memory or time is guaranteed. Define the same shock/common-suffix contract with $c = 1$. KV cache is part of state. A pure token-mediated arm must reconstruct a common-model cache from the canonical prefix and be distinguished from a persistent-cache arm. Report within-model paired effects rather than pooling raw LPIPS across models. Failure to repeat $H \downarrow$ would refute universal sign-pattern replication, even if incidence/consequence separation remains useful. Continuous diffusion or MAR with threshold-defined onset introduces a different estimand.

P3: non-saturated precision and coupling robustness. After establishing W8 granularity, select W6 or one alternative calibration method on development data and freeze it before held-out evaluation. W4 saturation is not primary generalization evidence. For another coupling, validate marginals and self-cases and report how h and paired discrepancy change. If effect signs or policy conclusions fail across predeclared valid couplings, report coupling dependence rather than universality.

Separate practical-value test. Retain any existing layer-by-time plan, but distinguish persistent W8 in one module, a one-time shock, and restoring one module to FP32 from an all-W8 baseline. Do not assume quantizing one module and restoring one from all-W8 are symmetric or additive. Compare local MSE, logit KL/TV, h alone, seed count and severity, simple early protection and same-cost random selection. Freeze a policy on development blocks, then evaluate actual restoration gain on entirely new class/seed blocks at matched parameter bytes or measured cost. Exclude post-hoc descendant features from online predictor inputs. If the method offers no added gain over these baselines, narrow the practical contribution. Separate cost/variance pilots from confirmation and choose sample size from effect size, power and multiplicity, without guaranteeing that a particular block count suffices. Any human study must distinguish blinded paired similarity from quality/preference and use representative samples; none is present here.

10 Conclusion

FDCA specifies how to join a natural approximation-induced token split to its conditional output discrepancy. Under a fixed Halton schedule, coordinatewise coupling and a one-step FP32-versus-W8/common-W8-suffix intervention, two CFG3 checkpoint panels support lower batch incidence but higher equal-block conditional and incidence-weighted LPIPS at target 0.5 than at 0.3. The weighted result is an average of blockwise products, and batch incidence also depends on commit opportunities. The theory supplies finite monotone-decoder bounds and an expectation identity, not a universal trend or a perceptual quality guarantee. Prospective prediction, precision selection and new-architecture validation remain open experiments.

Data, Code, Version and Citation

The public repository is `https://github.com/Munsik-Kim/fdca-visual-token`. Archived v1.0.0 remains associated with 10.5281/zenodo.21965747. This GitHub revision candidate is not a new release or Zenodo deposit. The author, ORCID and license scopes are preserved: original code is MIT; paper, figures, tables and included audit documentation are CC BY 4.0. Third-party weights, data and upstream code retain their own rights and are not redistributed.

The reanalysis script generates the numerical JSON/CSV, manuscript table inputs and vector/raster figures from the original compact tables. Its audit records input SHA-256 hashes, environment, RNG mapping and resampling rules. The version-aware paper build compiles this source with its updated bibliography; legacy frozen-paper verification remains distinct. Revision notes identify corrections, unresolved items and historical evidence boundaries.

A Post-hoc Robustness Supplement

These analyses were added after the original experiments. They are not the original preregistered analysis. All paired differences are $0.5 - 0.3$ within one panel. Table 9 uses 100,000 paired block bootstrap draws and linear quantiles, with ratio and split-population conditional means recalculated per resample. It is intentionally separate from the original 20k/higher headline intervals.

Table 9: Post-hoc natural-shock contrasts, with panel-specific n. CFG1 and W4 are overlapping sensitivity/stress subsets. The precise fixed RNG and input hashes are in the generated audit.

Panel	Metric	n	Δ	Post-hoc 95% CI
Tiny CFG3 W8	H	48	-0.083471	[-0.089371, -0.077090]
Tiny CFG3 W8	M_equal	48	0.020000	[0.017206, 0.022847]
Tiny CFG3 W8	R	48	0.001714	[0.001367, 0.002068]
Tiny W4 stress	H	24	-0.051112	[-0.065503, -0.037769]
Tiny W4 stress	M_equal	24	0.024676	[0.015529, 0.036123]
Tiny W4 stress	R	24	0.022024	[0.013202, 0.032775]
Small CFG3 W8	H	40	-0.071710	[-0.079636, -0.063458]
Small CFG3 W8	M_equal	40	0.020376	[0.017145, 0.023809]
Small CFG3 W8	R	40	0.001966	[0.001502, 0.002472]
Small CFG1 W8	H	16	-0.066112	[-0.073259, -0.058356]
Small CFG1 W8	M_equal	16	0.029132	[0.021132, 0.036801]
Small CFG1 W8	R	16	0.001686	[0.001045, 0.002269]

Table 10: Robustness diagnostics and the explicitly limited six-contrast Bonferroni family where indicated. Paired t, leave-one-block-out and trimmed summaries are descriptive sensitivity checks, not randomized causal inference.

Panel	Metric	Paired t 95% CI	Bonferroni CI
Tiny CFG3 W8	H	[-0.089837, -0.077105]	[-0.091408, -0.074753]
Tiny CFG3 W8	M_equal	[0.017079, 0.022921]	[0.016293, 0.023832]
Tiny CFG3 W8	R	[0.001352, 0.002077]	[0.001251, 0.002197]
Small CFG3 W8	H	[-0.080091, -0.063328]	[-0.082159, -0.060662]
Small CFG3 W8	M_equal	[0.016886, 0.023866]	[0.016101, 0.025094]
Small CFG3 W8	R	[0.001457, 0.002475]	[0.001355, 0.002668]

Panel	Metric	LOO range	10% trim
Tiny CFG3 W8	H	[-0.085237, -0.082607]	-0.084535
Tiny CFG3 W8	M_equal	[0.019461, 0.020522]	0.019718
Tiny CFG3 W8	R	[0.001645, 0.001766]	0.001682
Small CFG3 W8	H	[-0.073299, -0.070506]	-0.073246
Small CFG3 W8	M_equal	[0.019620, 0.020881]	0.019494
Small CFG3 W8	R	[0.001849, 0.002019]	0.001761

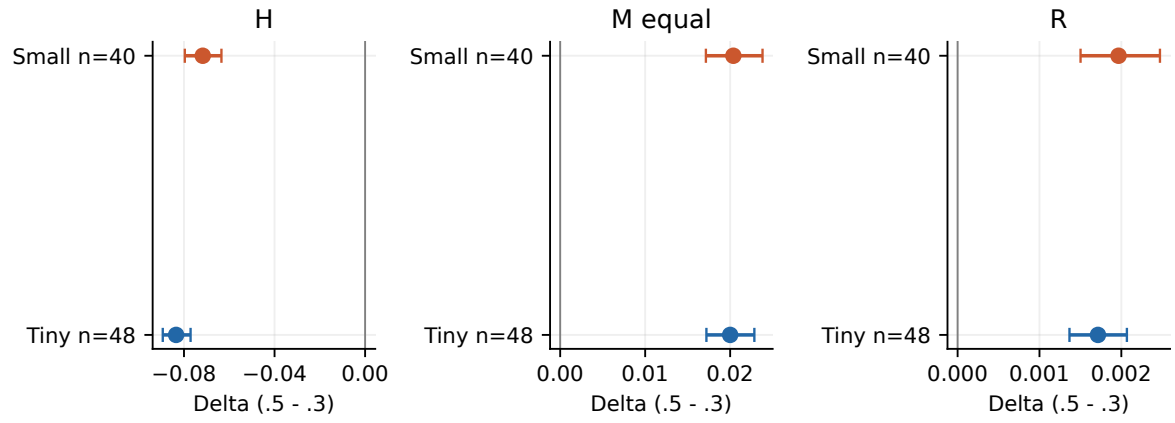


Figure 3: Post-hoc paired contrasts, 0.5−0.3, resampling complete block bundles. The displayed interval rule is 100k/linear; axes and panel counts distinguish probability from LPIPS metrics.

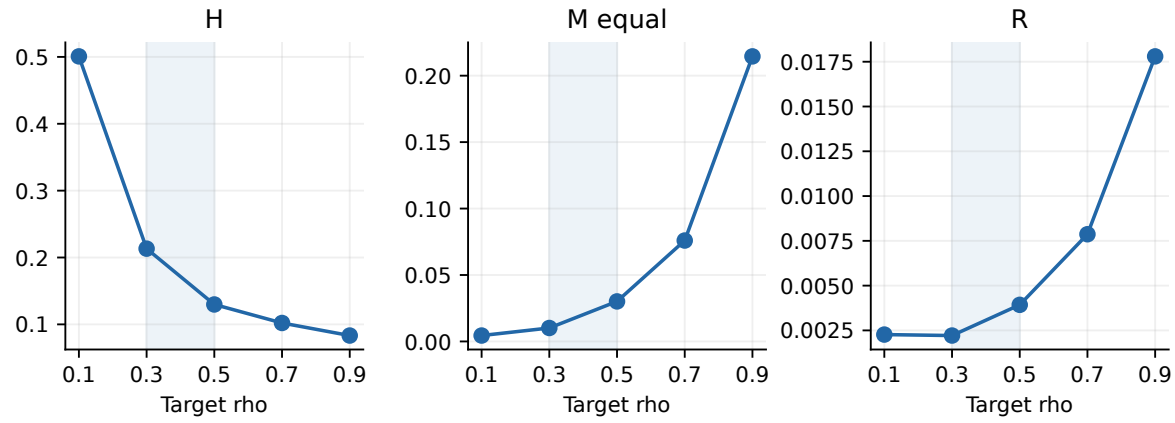


Figure 4: Full Tiny CFG3 W8 anchor profile ($n = 48$). All five target fractions are shown, including the final-flush 0.1 anchor. The mean of blockwise products R does not increase at every adjacent anchor. This full-anchor descriptive figure shows means without intervals; the selected 0.3/0.5 region is shaded, and adjacent contrasts are supplied in the generated tables.

B Historical Evidence and Proof Ledger

The original program progressed through T0 exact finite-state checks, P0 sampler inspection, P05R influence measurement, P1L controlled shocks, P1M incremental analysis, P2N natural seeds, P2A's incomplete audit, P2B's targeted closure, P2P perceptual measurements and P3S second-checkpoint replication. This order matters for selection: the P2P contrast followed earlier P2N findings; later new-block records are not equivalent to an externally timestamped preregistration.

The archived P3S report distinguishes 52 executed pytest categories from 60 generated audit assertions. It reports 1,429 run-manifest files, whereas the Full package contains 1,431 entries including two bookkeeping files. Its 1,040 paired-grid files include primary, matched, controls and the CFG1 subset; this count is not a count of independent blocks. The audit did not independently rehash excluded checkpoint bytes or rerun a Parquet pipeline in its environment. Likewise, current OOF MAE reaggregation is not independent model training.

The proof ledger in `docs/PROOF_LEDGER_v1.0.1.md` traces each theorem's assumptions and proof steps. The stable-set theorem is verified for its stated nonnegative bounded additive costs. The full-path decomposition is verified as a conditional-expectation identity with survival and zero no-split cost. The DAG result is verified conditional on the stated global hybrid closure and coordinate coupling; empirical local neural certificates and token-to-LPIPS transfer remain unproved. The ledger does not classify a familiar-looking statement as verified without those proof connections.

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